

Technical Specification for a new Electron Source and Linear Accelerator

1.8.27.1 Rev. 0

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1.0 PURPOSE

The Canadian Light Source (CLS) requires an Electron Source and Linear Accelerator (LINAC) to generate 250 MeV electrons at a repetition rate of 1 Hz. This document provides the vendors scope of supply and a set of performance and other parameters for the design and construction of the new Electron Source and LINAC.

2.0 SCOPE

CLS is planning to procure a replacement “turnkey” Electron Source and LINAC. CLS’s intent is that the successful vendor will supply all equipment required for a complete LINAC system, which would include, but not be limited to, the electron source, bunching section, accelerator sections, RF systems, vacuum system, diagnostic and control systems and cooling/heating systems.

3.0 GENERAL BACKGROUND

The CLS is a national research facility on the University of Saskatchewan campus in Saskatoon, Saskatchewan. The facility is a 3rd generation synchrotron light source with an electron beam energy of 2.9 GeV and current up to 500 mA, which produces high intensity beams of infrared, visible, ultraviolet and x-ray radiation.

The existing LINAC complex is built on a 1960s era 2856 MHz RF, normal conducting, 250 MeV electron linear accelerator that injects into a booster synchrotron that brings the beam up to an energy of 2.9 GeV, which in turn injects into the storage ring.

The existing LINAC is located in the basement tunnel with a klystron gallery on the ground floor above and operates asynchronously with the 500 MHz RF in the booster ring and storage ring. For standard injection, the LINAC operates in multi-bunch mode at a rate of 1 Hz but has also operated in top-up mode injecting a bunch train every few minutes. The LINAC is fed by a 220 kV DC electron source with a chopper and bunching sections to reach 13 MeV before injected into SLAC type travelling wave accelerating structures.

4.0 DEFINITIONS AND ABBREVIATIONS

ACH: Air Changes per Hour

ACIS: Access Control Interlock System

AHJ: Authority Having Jurisdiction

CAD: Computer Aided Design

CCG: Cold Cathode Gauge

CLS: Canadian Light Source, the working name of the facility

CSA: Canadian Standards Association

ECS: Energy Compression System

EPICS: Experimental Physics and Industrial Control System

FAT: Factory Acceptance Tests

FCT: Fast Current Transformer

FWHM: Full Width Half Maximum

GUI: Graphical User Interface

HV: High Voltage

IOC: Input/Output Controller, provides a software interface to control points in a system

ICT: Integrated Current Transformer

IP: Intellectual Property

ITP: Inspection and Test Plan

NDA: Non-Disclosure Agreement

LCW: Low Conductivity Water

LINAC: Linear Accelerator

LLRF: Low Level RF

OFC: Oxygen Free Copper

OPI: Operator Interface; provides the Human/Machine Interface to control points from the IOC

RF: Radio Frequency

RGA: Residual Gas Analyzer

RTD: Resistance Temperature Device or Resistive Thermal Device

SAT: Site Acceptance Tests

TCG: Thermal Capacitance Gauge or Thermal Conductivity Gauge

UHV: Ultra High Vacuum

VSWR: Voltage Standing Wave Ratio

WCM: Wall Current Monitor

X: Horizontal direction perpendicular parallel to electron beam axis

Y: Horizontal direction parallel perpendicular to electron beam axis

Z: Vertical direction

5.0 REQUIREMENTS

5.1 GENERAL REQUIREMENTS

5.1.1 Vendor Scope of Supply Summary

CLS intent is that the vendor will supply a complete “turnkey” Electron Source and LINAC system. The Electron Source and LINAC shall consist of, but not be limited to, the following major components supplied by the vendor:

- Electron Source
- Bunching sections
- Accelerator sections
- RF systems
- Vacuum systems
- Cooling/heating systems
- Diagnostic and control systems for all vendor supplied equipment
- Support structures and stands with adjustable height

- Lifting beams, jigs and rigging required to lift equipment.
- Specialized carts or wheel arrangements (if required) to move equipment within the facility
- Electrical power bussing, thermal sensors, tilt sensors, absolute encoders, motors, motor drivers, limit switches, motor junction box, control signal junction box, and wiring from motors and sensors to respective junction boxes.
- Mechanical fasteners and supports.
- Bakeout tapes and controllers for FAT (if required).
- Vacuum gauges, and pumps for FAT.
- Power supplies and high potential testing equipment for FAT.
- Vacuum pumps. The vendor shall design the vacuum pumping to meet the vacuum specification. The choice of pumps is subject to approval by the CLS.
- Vacuum gauges and the associated controllers. The vendor shall design the vacuum gauges required to meet the vacuum specification. The choice of pumps is subject to approval by the CLS.
- Technical Specialist to provide technical direction to CLS staff during the installation and commissioning of the vendor supplied equipment.
- Training to CLS staff on the operation of the vendor supplied equipment during commissioning stage.
- A schedule for the manufacturing, inspection, delivery, installation, commissioning and submittal of documents. This schedule shall include the estimated CLS manpower requirements for installation and commissioning.
- Documentation as specified within this document.

5.1.2 CLS Scope of Supply Summary

- Installation of the equipment under the technical direction of the vendor's technical specialists.
- Skilled technicians (Mechanical, Electrical, Instrumentation, etc) that will perform the hands on installation and commissioning work.
- The Electron Source and LINAC system will interface with several CLS systems that CLS will be responsible for. The interfaces and CLS systems are defined in further detail in the remainder of this document.

5.1.3 Scope of Supply Schematic

Figure 1 shows a schematic of the scope of supply with the CLS providing the laboratory space which will house the LINAC. The CLS uses an Energy Compression System (ECS) prior to injection of the electron beam into the booster due the narrow energy acceptance of the booster. The CLS will provide the ECS system and a bypass transfer line to reach the beam diagnostic lines. The CLS will provide the diagnostic lines for the SAT performance tests and will consult with the vendor supplying the LINAC on the requirements of the diagnostic lines.

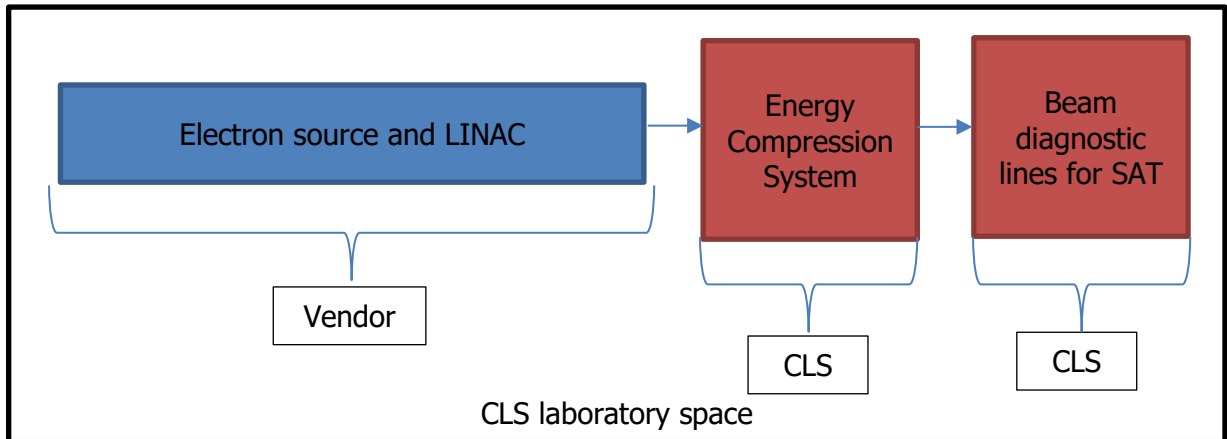


Figure 1: Schematic indicating the high level scope of supply of the LINAC system.

5.2 ELECTRON SOURCE AND LINAC SPECIFIC REQUIREMENTS

5.2.1 Performance Requirements

The following technical specification describes a turn-key electron linear accelerator. The LINAC shall meet the performance parameters given in Table 1. The LINAC energy is required to be 250 MeV. The LINAC will operate in single bunch mode, 1 A peak current, and a 2 to 136ns width multi-bunch mode. The repetition rate of the LINAC shall be 1 Hz to match the CLS booster synchrotron ring ramping rate of 1 Hz, with possible higher repetition rates for RF conditioning or beam commissioning. Proposed solutions to achieve these performance requirements are set out in Section 5 of this document and vendor's proposed solutions are subject to approval by CLS.

Table 1. LINAC Performance Parameters.

Accelerator Particles	Electron	
Energy	250	MeV
Beam (1 ns single bunch mode)	1	nC
Beam (2 to 136 ns multi-bunch mode)	3	nC
Center energy stability (pulse to pulse)	≤ 0.25	% (rms)
Energy Spread (rms)	≤ 0.5	% (rms)
Normalized Emittance (1s) (X or Y)	50	n mm·mrad
Frequency	2997.924	MHz
Sub-Harmonic Frequency	499.654	MHz
Repetition rate	1-10	Hz
Pulse to pulse beam position variation (rms)	0.2 mm	mm
Pulse to pulse beam angle variation (rms)	0.05	mrad

5.2.2 Electron Gun

The electron gun can generate electrons by using a thermionic cathode with a 80-100 kV accelerating DC voltage. A grid should also be used to modulate electron pulses when the system is triggered. CLS will provide the required signals from the CLS timing system including the 499.654 MHz signal and a variable 1 to 10 Hz LINAC trigger signal. The electron gun and its power and control system shall be easily replaced.

In single bunch mode, each single pulse from the gun should be shorter than 1 ns to fill one single 499.654 MHz bucket for the booster. Therefore the gun will produce 2A peak current with short pulses of 1 ns FWHM. In multi bunch mode, the gun will directly produce a chopped average 80 mA beam in 2-136 ns (adjustable).

For Top-up injection, the peak current, focusing and repetition rate can be adjusted.

Table 2. Electron Gun Requirement

Parameter	Value	Unit
High Voltage	-80 to -100	kV
Charge (1 ns single bunch mode)	2	nC
Charge (2-136 ns multi-bunch mode)	5	nC
Center energy stability	0.5	%
Energy Spread (rms)	0.5	%
Normalized Emittance	10.0	mm-mrad
Repetition	1-10	Hz

5.2.3 RF System

The expected LINAC RF system should be composed of a 499.654 MHz sub-harmonic pre buncher cavity that can pre-bunch electrons. Following this, a 2997.924 GHz single cell cavity acting as a main LINAC pre-buncher, will further compress the electron beam. Additionally, a 2997.924 MHz traveling wave structure buncher and several 3 GHz traveling wave structures will be installed to obtain proper quality beam. LLRF and RF power system shall provide sufficient phases and RF levels to be amplified to feed into the acceleration sections. The control and cooling system shall be designed to support the proper function of the RF system.

5.2.3.1 Sub-Harmonic Pre-Buncher (SHPB)

The sub-harmonic buncher is a standing wave cavity that is used to concentrate electrons longitudinally by modulating the electron velocities. The main parameters are shown in Table 3. The SHPB frequency is the same as the booster frequency that equals one sixths of the frequency used in the LINAC, which is 2997.924MHz. The advantage of this configuration, is to capture more electrons to be ramped in the booster, with good emittance and energy dispersion.

The sub-harmonic pre-buncher has to be temperature stabilized by the cooling water system. The vendor shall deliver a system with sufficient temperature stability to avoid active tuning during operation.

Table 3. 499.654 MHz Sub Harmonic Pre-Buncher Parameters

Parameter	Value	Unit
Frequency	499.654	MHz
Frequency Stability	$<5 \times 10^{-7}$	/day
Harmonic and Parasite Frequency	<-40	dB
Microwave Pulse Width	1.5~4 (depend on design)	μ s
Microwave Pulse Leading Edge	<100	ns
Microwave Pulse Jitter	$<\pm 5$	ns
Amplitude Stability	<1	%
Maximum Power	(depends on design)	kW
Repetition	1-10	Hz
Phase Adjustable Range	$>360^\circ$	degree

5.2.3.2 2997.924 MHz Pre-Buncher

The 2997.924 MHz pre-buncher is a single-cell copper cavity which makes another phase focusing after the sub-harmonic pre-buncher. Electrons going through the pre-buncher at the nominal phase of 0° will experience no energy gain. But the electrons 90° ahead will see the minimum gap voltage and be decelerated. The particles located at -90° get the maximum acceleration. After a drift distance, the electrons will be compressed into a smaller phase range. Table 4 lists the main requirements.

Table 4. 2997.924 MHz Pre-Buncher Parameters

Parameter	Value	Unit
Frequency	2997.924	MHz
Cavity Number	1	
Frequency Stability	$<5 \times 10^{-7}$	/day
Microwave Pulse Width	1.5~4 (depend on design)	μ s
Microwave Pulse Leading Edge	<100	ns
Microwave Pulse Jitter	$<\pm 5$	ns
Amplitude Stability	<1	%
Maximum Power	(depend on design)	kW
Repetition	1-10	Hz
Adjustable Phase Range	$>360^\circ$	degree
Working Temperature Tolerance	± 0.1	degree

5.2.3.3 2997.924 MHz Buncher

The buncher is composed of many beta matched traveling wave cavities, working in $2\pi/3$ -mode at 2997.924 MHz. The buncher is used as a longitudinal compression stage. Besides longitudinal beam focusing, the buncher increases the beam energy from 80-100 keV to 3-4 MeV. After a drift, the beam is injected to accelerating section 1.

The cell sizes, cavity number, maximum accelerating field, shunt impedance, energy gain, filling time, and input RF power are flexible, and should be determined by the vendor. Some buncher parameters are listed in the table 5.

Table 5. 2997.924 MHz Buncher Requirement

Parameter	Value	Unit
Frequency	2997.924	MHz
Mode	$2\pi/3$	
Velocity $\beta=v/c$	about 0.5 to 1	c
VSWR	1.05	-
Band Width (VSWR 1.05:1)	4	MHz
Quality factor (Q)	10200	-
Material	OFC	
Temperature Tolerance	± 0.1	degree

5.2.3.4 Accelerator Sections

CLS preference is to use five identical normal conducting copper accelerating structures. These sections will increase the energy up to 250 MeV with the required reliability of the system. The traveling wave sections work at 2997.924 MHz frequency (S-Band). For good efficiency, the $2\pi/3$ mode and constant field gradient is foreseen. The accelerating systems are operated at modest field and provide sufficient shunt impedance that a beam energy of 250 MeV can be reached. The accelerating structures have to be temperature stabilized by the cooling water system. The vendor shall deliver a system with sufficient temperature stability to avoid active tuning during operation.

Table 6. Accelerator Section Parameters

Parameter	Value	Unit
Frequency	2997.924	MHz
Type	Constant field gradient	
Mode	$2\pi/3$	
Structure length	3 to 5	m
Number of cells	depends	
Filling Time	depends	μs
Q-factor	>13500	-
Group velocity	depends	%c
Shunt Impedance R_{sh}	55.7 to 71	M Ω /m
R_{sh} expected average	52	M Ω /m
Velocity $\beta=v/c$	1	
Input RF Power	15 to 20	MW
Temperature Tolerance	± 0.1	$^{\circ}\text{C}$

5.2.3.5 Low Level RF System

CLS will provide a master 499.924 MHz RF signal for the LINAC. The low power RF system divides this master signal into 3 branches. The first branch may be used to feed the pulse electronics for the gun (if applicable). The second branch is used to drive the 499.924 MHz sub-harmonic pre buncher. The third branch is used for the 2997.924 MHz system. The 2997.924 GHz system consists of four adjustable (amplitude and phase) low level branches. One for the 2997.924 GHz pre-buncher amplifier, and one for each power plant. The frequency is generated from the 499.924 MHz master signal using a frequency multiplier (x6).

The 499.924 MHz signal for the 499.924 MHz sub-harmonic pre-buncher is amplified up to several hundred Watts to feed the related components. The 2997.924 GHz signal is amplified up to several hundred Watts to drive the klystrons.

Table 7. LLRF System Distribution

CLS	Vendor			
	499.924 MHz	2997.924 MHz		
	Gun (if applicable)	LINAC Phase Shifter Control	LINAC Attenuator Adjustment	Component
	Sub-harmonic Cavity			
499.924 MHz 13 dBm RF signal	Frequency Multiplier(X6) to produce 2997.924 MHz RF signal and Amplifier	>360° phase shifter	>20 dB Attenuator	Klystron 1- Prebuncher
		>360° phase shifter	>20 dB Attenuator	Klystron 1 - Buncher
		>360° phase shifter	>20 dB Attenuator	Klystron 2 - Section 1
		>360° phase shifter	>20 dB Attenuator	Klystron 3 - Section 2
		>360° phase shifter	>20 dB Attenuator	Klystron 4 - Section 3

5.2.4 RF Power System

5.2.4.1 Klystrons

The RF power system consists of several modulators and klystrons. The klystron peak power could be limited by our modulator room height (3.2 m, see drawing 0047250 and 0047251). We are currently using six Toshiba E37300 klystrons which provide 20 MW RF power for each LINAC section. Toshiba are the preferred supplier for the klystron systems.

5.2.4.2 Preamplifiers and 499.924 MHz Amplifier

Each klystron needs a preamplifier to be driven. A solid state amplifier may be the best solution. A possible preamplifier specification is listed in table 7.

Table 8. Preamplifier Specification

Frequency range	2.998 GHz $\pm$ 5 MHz
Output power	400 Watts pulsed
Class of operation	AB
Pulse length	up to 20 μ s
Repetition rate	up to 10 Hz
Duty cycle	0.10%

The 499.924 MHz signal for the 499.924 MHz sub-harmonic pre-buncher is amplified up to several hundred Watts to feed the related components.

5.2.5 Modulators

The modulators will be used for driving klystrons. Each modulator provides pulsed high voltage for the klystrons. ScandiNova solid status K2 modulators are preferable. We have been using them for seven years. The K2 modulators are reliable, require less maintenance, and are stable.

5.2.6 High RF Power Distribution

WR284 waveguides should operate in high vacuum. One ion pump should be located directly at the klystron output window. The waveguide system vacuum will be separated from the LINAC vacuum by means of ceramic windows. There shall be an ion pump on either side of each ceramic window. Each klystron shall have a waveguide bidirectional coupler that is used for monitoring forward and reflected RF power at the output of the klystron and at the input to the structures. Several power splitters may be used to distribute the RF power to RF components. Waveguide phase shifters may be used if applicable (depends on vendor's design). Dummy loads must apply be applied at all accelerator sections.

5.2.7 Focusing and Steering Magnets

The electrons from the thermal cathode are imparted angular momentum. RF defocusing and space charge effects naturally defocus the beam. Solenoids and quadrupoles must be used to keep the beam in a given area. The solenoids are used in the low energy region to ensure radial symmetry and to avoid large amplitude oscillations in the transversal plane of the bunching section. The quadrupoles, between the accelerating sections, will be used to focus the beam at higher energy. To compensate for misalignment and steering effects of the magnets, steering magnets are used between LINAC acceleration sections.

5.2.7.1 Solenoids

The first solenoid of the system may be placed right after the electron gun. This solenoid should be installed on an adjustable support to adjust the solenoid on the beam axis of the LINAC.

The water cooled solenoids are installed with the buncher to insure good field homogeneity through the focusing channel avoiding high emittance growth of the low energy beam in the solenoid channel.

5.2.7.2 Quadrupoles

A quadrupole triplet should be installed at the 50 MeV region. For the 250 MeV LINAC, an additional three quadrupole triplet should be at the 150 MeV region for matching the beam. All quadrupoles are identically wired, but the power supply connection should be done correctly to alter the focusing plane. To minimize the effect of quadrupole kicks, the magnetic axis will be aligned with respect to the beam line to within 0.05 mm.

5.2.7.3 Steering magnet

Each steerer can move the beam by a few (2-4) millimeters per meter. Due to the fact that steerers with iron yokes cannot be used within solenoids, two different types of steerers should be used. The Helmholtz type uses just two air coils to produce the dipole field and can therefore be used within the solenoids. Window type steerers will be installed in the higher energy area.

5.2.8 Beam Diagnostic System

5.2.8.1 Several Diagnostic Parts

The function and resolution for several electron beam diagnostic elements are listed below.

5.2.8.2 Faraday Cups

50 Ohm matched Faraday Cup can measure the charge and the pulse structure with a fast oscilloscope (3 GHz real time). The Faraday Cup should be moved by a pneumatic drive.

5.2.8.3 Wall Current Monitor (WCM)

Broadband wall current monitors (WCM) will be used to measure beam current with the bandwidth > 3 GHz.

5.2.8.4 View Screen with CCD Camera (Optical Monitor)

The image given by the screens will be analyzed by a CCD camera (available on the Ethernet control network) and will measure relative beam position and profile.

5.2.8.1 Fast Current Transformer (FCT)

A fast current transformer (FCT) can be used to measure beam current with bandwidth > 3 GHz. The FCT is functionally the same as the WCM.

5.2.8.2 Integrating Current Transformer (ICT)

The Integrating *Current Transformer (ICT)* is a capacitive shorted *transformer* and a fast read out *transformer* in a common magnetic circuit, designed to measure the charge in a very short pulse with high accuracy.

Table 9. LINAC Diagnostic Element Requirements

LINAC Diagnostic Element Requirements		
Parameter	Detector	Resolution
Beam waveform (single & multi-bunch)	FCT or WCM	3 GHz BW
Charge	ICT or Faraday Cup	2%, 3 GHz BW
Energy and energy spread	Dipole and detector	5x10E-4
Relative position	Optical Monitor + CCD Camera	50μm
Emittance	Wire scanner or equivalent	15%

5.2.9 Diagnostic Element Distribution in LINAC

5.2.9.1 *At Gun Exit*

Beam emitted from the gun is very important for operators. They need to know beam current, pulse waveform (single bunch, multiple-bunch), Charge, beam profile, and position. FCT or WCM should be installed here to measure the beam current and waveform. A Faraday cup or ICT can measure the charge. An optical monitor with a CCD Camera can measure the beam position and profile.

5.2.9.2 *At Sub-Harmonic Pre-Buncher Exit*

Only the beam position and profile should be measured. An optical monitor with a CCD Camera should be installed here.

5.2.9.3 *At the End of Buncher*

We need to know Charge, beam profile and position. A Faraday cup or ICT can measure the charge. An optical monitor with a CCD Camera can measure the beam position and profile at this location.

5.2.9.4 *At the End of Sections*

At each end of the accelerator sections, the operator should be able to adjust the steering magnet. We need an optical monitor with a CCD Camera installed at these locations.

5.2.9.5 *At the End of LINAC*

Beam quality is very important at this location. Operators need to know beam current, pulse waveform (single bunch, multiple-bunch), charge, beam profile and position, beam energy, and energy spread. The first magnet at the ECS (Energy Compress System) can be used to measure beam energy and energy spread. The beam position and profile can be seen on the view screen. We also can use a beam-stop to measure the current and pulse waveform.

All elements here already exist. It is not necessary to install new ones.

5.2.9.6 *List New Diagnostic Elements in the New LINAC*

- One WCM or FCT (after the electron source)
- Two removable Faraday Cups (after electron source and after bunching section)
- 6 removable screens including cameras after the electron source, SPB, bunching section and after the first two accelerating sections.
- Additional removable screens including cameras after the accelerating sections or after a pair of accelerating sections.

Table 10. LINAC Diagnostic Elements and Locations

Location	Parameters Measured
Electron Gun Exit	Beam current waveform(single bunch, multiple-bunch), Charge, Profile and position 【FCT, OP+CCD Camera】
SHPB Exit	Position 【OP+CCD Camera】
Buncher Exit	Beam current waveform (single bunch, multiple-bunch), Charge, Profile and position 【FCT, OP+CCD Camera】
SPB Exit	Position 【OP+CCD Camera】
Each Section Exit	Position 【OP+CCD Camera】
End of LINAC	Beam current waveform, profile, position, energy, energy spread, emittance 【FCT, Dipole, OP+CCD Camera】

5.3 SPATIAL CONSTRAINTS

- 5.3.1** The coordinate system is described as follows: +X following the electron trajectory, +Y outboard, and +Z up. See CLS Drawing 0047250.
- 5.3.2** CLS Drawing 0047250 shows multiple views of the free space available in the LINAC hall. In the above drawing, the fixed structures are shown in black and the removable structures are shown in green. The electron beam position is shown at an elevation of 1219 mm from the floor.
- 5.3.3** There is a run of HVAC ducting above the current LINAC. This ducting is shown in aqua color on CLS Drawing 0047250. The ducting may be considered removable if necessary. If the Vendor requires any changes to the existing ductwork, CLS shall be notified at the preliminary design review stage.
- 5.3.4** The ceiling of the LINAC tunnel has one wave guide penetration to the modulator testing room and three wave guide penetrations to the modulator room that is used for LINAC operations. The general positions of these features are outlined in CLS Drawings 0047250 and 0047251. Detailed survey location of these features will be provided to the successful Vendor.
- 5.3.5** The existing CLS modulator room is located above the LINAC hall at ground elevation or 6.4 m above the LINAC floor level. The footprint of the modulator room and test klystron room is super-imposed as a dashed magenta line on CLS Drawing 0047250. Reference CLS Drawing 0047251 for further details on these rooms. Modulators for the new LINAC system shall be located in the existing CLS modulator room. Modulators should be placed within the room such that any individual modulator can be removed/ replaced without moving adjacent modulators.

- 5.3.6** There are a number of spatial constraints for the new LINAC/Gun sections in terms of how they will be delivered to our existing LINAC hall. The crane well down to the LINAC hall has the tightest spatial clearances. No section or component should be larger than 4.0 m long X 0.8 m wide X 1.6m high in order to fit down the crane well horizontally. It is possible to accommodate longer components if the load is tilted while going down the crane well. If the load is tilted by 60°, a component may be up to 5.5 m long X 0.8 m wide X 1.5 m high. Any deviations above these values must be done with approval by CLS. The Vendor shall submit maximum sizes for all loads to be transported down the crane well at the preliminary design review stage.
- 5.3.7** Drawing 0047250 shows the locations of existing service trenches in the floor. These are available for LINAC related services but it should be noted that they are prone to water accumulation.

5.4 SUPPORT STRUCTURE

- 5.4.1** The electron gun section and each accelerator section must be supplied with individual support structures.
- 5.4.2** The support structures shall be designed and fabricated in accordance with Canadian Light Source Requirements for Design and Fabrication of Structural Supports, CLS Document 8.8.29.1.
- 5.4.3** The height of the electron beam shall be 1219 mm from the floor.
- 5.4.4** The support structure should accommodate +/- 10mm of manual adjustment of the components in X, Y, and Z. All adjustment screws should have precision threads with <2.5 mm adjustment per turn. The design shall have the adjustment mechanisms located in the center of their respective travel range to allow for equal movement in all directions during installation and alignment.
- 5.4.5** Support structures should be provided for heavier ancillary components such as ion pumps.
- 5.4.6** The design of each section shall include detachable wheel units or a cart in order to move the sections into place within the LINAC hall. One set of wheel units or one cart may be used for all the sections as they will be moved in sequence.
- 5.4.7** There are a number of service trenches and sump pits in the existing LINAC hall floor. See CLS Drawing 0047250. Support structures for the individual LINAC/gun sections shall straddle any nearby trenches or pits. No support structure can rest upon a trench or pit cover. Support structures must not prevent the removal of covers for the existing service trenches and sump pits.
- 5.4.8** Support structures shall allow for a 40mm grout allowance with the floor as per CLS Document 0.4.1.2. Design Specification for Precision Aligned Components.
- 5.4.9** Materials used that are subject to surface corrosion shall be anodized, passivated, or painted wherever possible (see CLS Document 8.8.29.1).

5.4.10 All sub-assemblies shall be stress relieved before final machining to minimize material creep (see CLS Document 8.8.29.1).

5.4.11 Moving and Lifting of Equipment at CLS. The design and vendors scope of supply shall include:

- The Vendor shall provide a lifting design drawing and plan to CLS for each LINAC/gun section. For sections longer than 4.0 m, the Vendor's lifting design shall allow the section support structure to tilt up to 60° as the load is lowered down the crane well. The design drawings and plans are subject to CLS review and approval. The shipping crates for each section shall not be used in the lifting design or component lifts.
- The lifting fixtures and supply of any hardware needed to lift the device with CLS cranes. All fixtures and hardware shall be approved for use in Saskatchewan by a Professional Engineer registered to practice in the Province of Saskatchewan.
- The center of gravity shall be clearly shown on each section and indicated in the drawings.
- The overhead crane for the LINAC crane well has a capacity of 5 tons (4536 Kg). The combined mass of each section and the associated lifting fixture shall be kept below this value.
- The lift points on the each section shall be positioned above the center of gravity of the device.
- Any rigging used to lift an individual section shall use engineered lifting points with shackles. "Basketed" lifting with slings wrapped around components shall not be used.
- All rigging supplied by the vendor must be approved for lifting purposes and clearly labeled and/or marked in English with the load rating and related information.

5.5 SURVEY FIDUCIALS

5.5.1 There shall be at least 8 survey fiducial locations on each LINAC/gun section. At least 4 locations shall be on the elevation of the electron beam centerline and 4 locations adjacent to the manual adjustment screws. Final positions to be approved by CLS.

5.5.2 The CLS survey equipment is setup on the +Y side of the LINAC. All fiducials must be visible from this vantage point.

5.5.3 CLS uses 38.1 mm diameter sphere mounted retro reflectors. Target seats shall be constructed as per CLS Document 0.4.1.2.

5.5.4 A network of survey nests exists in the LINAC hall. A map of these nests referenced to the CLS global survey network origin will be provided to the successful vendor.

- 5.5.5** The vendor shall provide the measured location of the electron beam relative to the fiducial points at the FAT stage for CLS review.

5.6 VACUUM SYSTEMS

- 5.6.1** The vacuum systems shall be designed to achieve an ultra-high vacuum. Vacuum pressure shall be $<1 \times 10^{-8}$ torr with the electron beam present. The vendor shall provide to CLS for review and approval a RGA spectrum up to molecular weight of 100.
- 5.6.2** The vacuum systems shall be fabricated in accordance with the CLS High Vacuum Design Specification 8.4.33.1 and the CLS Vacuum Component Fabrication Technical Specification 8.8.33.4.
- 5.6.3** The Vendor shall follow the CLS Component Cleaning Technical Procedure 8.7.33.1. Vendor may propose alternative cleaning procedures for CLS review and acceptance.
- 5.6.4** The electron gun system shall be the first vacuum section. Each subsequent accelerator section shall define the remaining vacuum sections. Each vacuum section shall be isolated from the adjacent sections by an all metal UHV gate valve (VAT series 48, 24VDC, with position indicators, preferred).
- 5.6.5** All ion pumps shall be selected from standard Varian, Agilent, or SAES sizes and shall be subject to approval by the CLS. A combination of ion pumps and NEG pumps is also allowed. The final pump selections shall be approved by the CLS.
- 5.6.6** Each vacuum section shall be equipped with cold cathode gauges (CCG) thermal capacitance gauge (TCG) and roughing port. The final choices shall be approved by the CLS.
- 5.6.7** There shall be a welded bellows between each adjacent vacuum section to allow for minor adjustment of one section relative to another. The size and length of the bellows shall be determined by the Vendor and approved by CLS.
- 5.6.8** The existing interface point to the CLS LTB transfer line is shown on CLS drawing 0047250. This existing section starts with a 6" CFF all metal gate valve. The Vendor must supply a rotatable 6" CFF bellows to connect the last accelerator section to the existing CLS section.
- 5.6.1** Each vacuum section shall have a helium leak rate $QL < 1 \times 10^{-7}$ Pa l/s. The vendor shall adhere to CLS Vacuum Component Leak Test Technical Procedure 8.7.33.2.
- 5.6.1** The vendor shall supply a bake procedure for any systems that require baking.

5.7 COOLING/HEATING WATER

5.7.1 Vendor to design and supply cooling/heating water systems required for stable operation of the new Electron Source and LINAC system, including the modulators. The control systems for the cooling/heating water systems are included in the vendor's scope of supply.

5.7.2 Design drawings and documentations shall include, but are not limited to the following:

- Process flow diagrams
- floor plans
- sections views
- Details of installation
- Control parameters and sequence of operations
- System and equipment literature, specifications and datasheet Including product characteristics, performance criteria, and limitations
- Manufacturer's installation instructions
- Maintenance requirements and manuals
- Warranty certificates

5.7.3 All LCW system shall meet 0.2.1.1 CLS Naming and Numbering Convention.

5.7.4 CLS supplies a primary cooling water (LCW) system to the LINAC service area operating at the following parameters:

- Supply temperature: $26 \pm 1^{\circ}\text{C}$
- Supply Pressure: maximum 55 psi
- Conductivity level: $<6 \mu\text{S/cm}$
- Cooling capacity: 20 kw
- Water flow rate: 20 GPM

- 5.7.5** CLS is capable to supply primary cooling water (LCW) system to the Modulator test room (shown on CLS Drawing 0047250, just next to the modulator room) or to the mechanical room 2001 (right above the modulator room) operating at the following parameters:
- Supply temperature: $26 \pm 1^{\circ}\text{C}$
 - Supply Pressure: maximum 15 psi
 - Conductivity level: $<6 \mu\text{S/cm}$
 - Cooling capacity: 70 kw
 - Water flow rate: 75 GPM
- 5.7.6** CLS supplies LCW re-fill to both LINAC hall, modulator test room and mechanical room 2001.
- 5.7.7** Vendors shall supply secondary closed LCW cooling/heating system, [e.g., system to be separated via heat exchanges (for systems with cooling) or expansion tanks (for systems with heating only) from the primary cooling/heating/refill system and will not opened to atmosphere during operation.]
- 5.7.8** Each section shall have an independent cooling/heating system skid with controls.
- 5.7.9** Modulators, klystrons, electromagnets and waveguides shall have a stand-alone cooling/heating system with controls.
- 5.7.10** All pressure equipment and components shall meet the following codes and standards:
- American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, 2017
 - American Society of Mechanical Engineers (ASME) Pressure Piping Code, B31.3, 2016
 - Saskatchewan Boiler and pressure vessel Regulations, 2017
 - Saskatchewan Boiler and Pressure Vessel Act, 1999
- 5.7.11** Each system shall have two pumps both of which are capable to meet 100% of the system flow requirements. One pump will operate and the other will be a stand by pump. The design will permit one pump to be removed for maintenance while the other pump continues to operate.
- 5.7.12** System shall have inline filtration (one micro absolute bag filter) to maintain system cleanliness and capture particulates.
- 5.7.13** System shall equip with side stream filtration (not less than 3% of system circulation) including resin mix bed to maintain conductivity level, carbon filter or UV light for a control of organic grow and a sediment filter to capture particulates/loose resin.

- 5.7.14** A vacuum degasification system shall be supplied to remove dissolved oxygen and maintain a system dissolved oxygen level of ≤ 20 ppb.
- 5.7.15** System shall be equipped with low velocity vortex air separator to remove entrained air from water.
- 5.7.16** System heat exchangers shall be plate to plate heat exchanger.
- 5.7.17** System heaters should be electric immersion heaters, 208V, single phase and CSA certified.
- 5.7.18** System expansion tanks shall be diaphragm or bladder expansion tanks.
- 5.7.19** System mixing valve shall use fast reaction electric actuators, not pneumatic actuators, if possible.
- 5.7.20** System return temperature shall be < 60 °C to avoid system classification as a pressure vessel/piping.
- 5.7.21** System supply temperature stability shall be sufficient to maintain proper stable operation of the LINAC.
- 5.7.22** Cooling/heating water system components shall be able to withstand a minimum of two times the maximum operating pressure.
- 5.7.23** All pumps, including stand-by pumps, shall be equipped with a run hour meter.
- 5.7.24** All fittings are imperial, not metric.
- 5.7.25** Each component shall be equipped with isolation valves for maintenance.
- 5.7.26** The supply and return headers shall have a shut off valve immediately after the pump skid.
- 5.7.27** Pump inlet/outlet shall provide metal flex connectors (316 stainless steel convoluted hose with outer protective braid) for vibration isolation.
- 5.7.28** All components requiring cooling/heating shall be run in parallel unless flow requirements are similar and combined pressure drops are reasonable if run in series.
- 5.7.29** LCW systems shall supply measured readings of return pressure, return temperature and return flow rate, locally and remotely for each cooling/heating component.
- 5.7.30** Each system shall have only one measurement for supply pressure and temperature when all components within a system are run in series.

- 5.7.31** LCW systems shall be equipped with flow switches at the water return of each cooling/heating system. The flow switches shall be interlocked with system operation.
- 5.7.32** Water systems shall have low conductivity indicators to show good/bad conductivity level.
- 5.7.33** Water systems shall be equipped with flow adjustment devices /flow control valves or circuit setters for each system and component.
- 5.7.34** There shall be an automatic air release device at the highest point of each loop and a drain at the lowest point of each loop.
- 5.7.35** Water system piping designs shall have enough degrees of freedom to allow for accurate install and easy field fitting. This is achieved using the right combination of welded, Victaulic, Sanitary Quick-Clamp pro-press, and American Standard Swagelok joints. Nation Pipe Thread (NPT) joints are acceptable for pipe sizes smaller than 1.5in (however they are less preferable).
- 5.7.36** Connections at equipment or components shall be an "easy" type: CLS preference is Sanitary Quick-Clamp for pipe size greater than 1" and Swagelok for tube size not more than 1". Flange or Victaulic connections are acceptable for pipe size greater than 1". Nation Pipe Thread (NPT) connections are acceptable for pipe sizes smaller than 1.5" (however they are less preferable).
- 5.7.37** All Materials must be compatible with deionized water and must be radiation resistant when installed in LINAC hall.
- 5.7.38** All water system materials including equipment, components, valves, fittings etc. shall be 316/316L stainless steel or copper.
- 5.7.39** Piping shall be Schedule 10, 316/316L stainless steel seamless piping or tubing that conforms to ASTM A312/A213 or ASTM A269/A269M. Threaded piping shall be schedule 40 316/316L seamless stainless steel. Copper tubing shall be type L that conforms to ASTM B88.
- 5.7.40** Flex connections shall be provided when connecting to equipment and components (using rubber tubing, Parker Paraflex Nylon tubing or 316 stainless steel metal flex tubing).
- 5.7.41** Instrumentation located in radiation areas shall be radiation hardy. Electronics shall be located outside of radiation areas or encased in protective shielding.
- 5.7.42** Pump skids shall be located on top of a house keeping pad with spring isolators for vibration control.
- 5.7.43** Pump skids shall be equipped with containment pans when there is electrical equipment on the floor or sub-floor.

- 5.7.44** All water system piping shall be supported from the ceiling, walls, or tele-posts using Unistrut cush-a-clamps. Pipe hangers shall be spring isolated.
- 5.7.45** Piping shall keep out of crane coverage areas and consider line of site for survey of alignment.
- 5.7.46** All Stainless Steel welding shall be Tungsten Inert Gas (Tig) welding in accordance with CSA W47.1-03.
- 5.7.47** All Stainless Steel welds shall be pickled in accordance with ASTM A380.
- 5.7.48** No direct water-to-vacuum joints (as welded, brazed, or mechanically assembled) can be used. This means joints within a water line, or any other pressurized line, must be situated outside of the HV or UHV vacuum volume or vented to atmosphere or to a secondary vacuum volume called an "airguard".
- 5.7.49** When bending welded tubing, the bend radius must not be less than the minimum bend radius for the tube diameter.
- 5.7.50** The components cooling channels must be pressure tested to a minimum of 1.5 times the maximum working pressure. They must also be cleaned, drained thoroughly, and capped before shipment to CLS.
- 5.7.51** All interface points shall be clearly labeled as supply or return.
- 5.7.52** Components shall be labeled with the following information:
- cooling/heating load
 - flow rate
 - Inlet temperature and temperature stability required
 - Maximum allowable working pressure
 - Pressure drop

5.8 ROOM HEATING VENTILATING AND AIR CONDITIONING

5.8.1 CLS Air handling unit AHU2001-04 is located in mechanical room 2001, right above the modulator room. It supplies conditioned air into the LINAC hall and LINAC service area with the fan running at low speed to maintain the following parameters in the rooms:

- Temperature: 20-26 °C
- Humidity: no humidity control
- Air change per hour: 19.5 ACH @ high speed & 9.1 ACH @ low speed
- Cooling capacity: 95kw @ high speed/45kw @ low speed
- Heating capacity: 244kw @ high speed & 114 kw @ low speed with steam heating coil
- Flow rate: 10,909m³/h (6417cfm) @ high speed & 5,105m³/h (3003cfm) @ low speed

5.8.2 The existing ventilation system is a 100% outdoor air system with a supply air duct located overhead of the LINAC hall and the mechanical service area. See DWG 0047250. The air moves through these areas and exhausts from a vent stack in switchyard to outside.

5.8.3 The ducting system can be relocated/modified by CLS to fit the new LINAC installation if required.

5.8.4 Vendor shall supply the following information/requirements:

- Cooling/heating load from all vendor supplied equipment.
- room temperature and stability required
- room humidity control if required
- Room air changes per hour if required
- Room ventilation rate/make-up air rate if required

5.9 PNEUMATIC AIR SUPPLY

5.9.1 CLS is capable of pneumatic air supply from the central compressed air plant at the following parameters:

- Supply pressure: 689.5 kPa (100 psi)
- Moisture/humidity level: air dew point to be @ -40 °C at the central air dryer
- Central particulate filtration: 1 micron
- Central coalescing filtration: 0.01 micron

5.9.2 Pneumatic devices shall be compatible with CLS clean and dry instrument air supplied from the central compressed air plant.

5.9.3 Air consumption shall be conserved and used only to actuate devices.

5.9.4 Components shall be designed to be able to operate with supply air pressure as low as 483 kPa (70 psi).

5.9.5 Vendor shall supply the following information/requirements:

- Maximum and minimum air supply pressure required
- Air consumption requirements.

5.10 FIRE PROTECTION

5.10.1 Fire suppression and alarming shall be the responsibility of CLS.

5.10.2 Vendor shall identify the type of fire suppression system, if required, to protect Vendor supplied equipment.

5.11 CONTROL SYSTEM

5.11.1 The vendor shall supply an integrated control system for the Electron Source and LINAC equipment, including the cooling/heating water skids. The CLS prefers a turn-key control system that is compatible with CLS standards. Final design is subject to CLS approval.

5.11.2 The CLS uses a distributed control system as a high level interface to hardware. It is used for reading feedback variables and setting control set points over the CLS network. This permits us to access the control parameters of CLS hardware internal to CLS but at a remote location with respect to the hardware. All controls must be EPICS compatible. The vendor shall supply EPICS drivers for all control components. The CLS currently uses EPICS R3.14.12 running on LINUX.

5.11.3 The vendor shall provide the low-level control system software and a high level GUI. The GUI should be EDM or Control System Studio. The GUI shall be a user interface for exposing/setting/loading/running hardware monitors, control variables, scripting commands, faults statuses, operation sequence, system set commands, system reset commands, clearing latches, and clearing faults. EPICS software and operator screens shall meet 0.2.1.1 Rev 8 CLS Naming and Numbering Convention. Equipment names will be provided by the CLS after design is approved. Any deviations from these standards are subject to approval by the CLS.

5.11.4 Source code shall be provided for all EPICS components. Documentation and comments must be in English.

5.11.5 Vendor must verify control computers are acceptable to CLS before final acceptance. Substitutions must have CLS acceptance.

5.12 ELECTRICAL GENERAL

5.12.1 Submittals

The Vendor shall provide a list of electrical service requirements for all equipment, including AC supply voltage, current, overcurrent protection, motor FLA, and other information as required to establish electrical services.

The CLS is responsible to design, supply and build the electrical services, including distribution panels, breakers, supply cables and power connection of Vendor equipment.

All subcontracted components or work shall be clearly identified in the Vendor's proposal.

Vendor submittals shall include:

- Schematics
- Wiring diagrams
- Interconnection diagrams
- Shop drawings
- installation details of equipment indicating proposed location, layout and arrangement of control panels, accessories, piping, ductwork, and other items that must be shown to ensure coordinated installation
- clearances for operation, maintenance, and replacement of operating equipment devices.
- MTBF of individual critical components

5.12.2 Quality Control and Certification

All equipment shall be in compliance with the applicable electrical standards in Saskatchewan, including C22.1-18 Canadian Electrical Code, Part I (2018), Saskatchewan Interpretations (2015), Saskatchewan Electrical Inspection Act, E6-3, with special attention to section 18 "Manufacture, Sale, etc." of electrical equipment in the Electrical Inspection Act. This requirement may be satisfied either by certification to the relevant CSA standard by an authorized inspection agency, or by special inspection (SPE-1000) carried out by an authorized inspection agency. See the Saskatchewan Interpretations pages 17-21 for a complete list of recognized certification labels and special inspection labels in Saskatchewan. If it is not feasible to obtain a certification mark or special inspection mark prior to delivery, a special inspection can be arranged at the Buyer's premises, however any additional costs or equipment modifications required to obtain the field inspection shall be at the Supplier's cost.

Equipment shall be plainly marked in a permanent manner, where the details will be readily visible after installation without use of a tool, with the following:

- a) Manufacturer's Name;
- b) Model;
- c) supply voltage or voltage range;
- d) frequency or frequency range;
- e) number of phases, unless intended for single-phase use only;
- f) input in amperes, watts, or volt-amperes;
- g) for multi-voltage equipment, all rated voltages and current ratings;
- h) required caution markings;
- i) horsepower rating of motor(s) used within the equipment, where the motor nameplate(s) is not visible during inspection or installation;
- j) motor full-load current rating where motor overload protection is intended to be separately provided during installation;
- k) enclosure-type classification where intended for other than indoor or dry location usage; and
- l) for electrical equipment required to interrupt current and rated 750 V and less, interrupting ratings in accordance with applicable Part II Standards.

Refer to the Model Code for the field evaluation of electrical equipment, CSA SPE-1000-13, for the complete list of requirements for field certification in Saskatchewan.

5.12.3 Labelling and Identification

Equipment shall be identified in accordance with the CLS Naming and Numbering Convention. Equipment names will be provided by the CLS after design is approved.

5.12.4 Field Wiring

The Vendor shall supply proposed wiring diagrams for the field wiring of equipment. The CLS will be responsible to supply approved drawings and permits to the Authority Having Jurisdiction (AHJ) to meet code requirements.

The CLS is responsible for supply and installation of field wiring for power, control and data.

5.12.5 Enclosures

All electrical enclosures shall have an ingress protection rating of IP52, or better. Enclosures at ground elevation (not in the LINAC hall) must have an ingress protection of IP54, or better and shall not be damaged if subjected to water discharge from a fire suppression (sprinkler) system.

The maximum internal air temperature of electrical equipment enclosures shall be 40 degrees Celsius or lower.

5.12.6 Care, Operation and Startup

Vendor shall perform tests in accordance with this section as noted in SAT.

Vendor shall instruct CLS Representative and operating personnel in the operation, care and maintenance of systems, system equipment and components.

Operating instructions to include following:

- Wiring diagrams, control diagrams, and control sequence for each principal system and item of equipment.
- Start up, proper adjustment, operating, lubrication, and shutdown procedures.
- Safety precautions.
- Procedures to be followed in event of equipment failure.
- Other items of instruction as recommended by manufacturer of each system or item of equipment.

Vendor to arrange and pay for services of manufacturer's factory service engineer to supervise start-up of installation, check, adjust, balance and calibrate components and instruct operating personnel.

Provide these services for such period, and for as many visits as necessary to put equipment in operation, and ensure that operating personnel are conversant with all aspects of its care and operation.

5.13 SAFETY PERFORMANCE

5.13.1 Access Control Interlock System Interface

The Access Control Interlock System (ACIS) is used for personnel protection. The ACIS ensures the shielded LINAC areas are cleared of personnel before radiation sources or high voltage hazards may be enabled. There is a potential for loss of life or significant injury if the LINAC does not shut off when required.

The CLS has an existing ACIS system for the LINAC areas, which is intended to be reused for the new LINAC. The CLS will modify the ACIS to suit the vendor's proposal for any identified radiation sources or high voltage hazards and changes required to LINAC shielding or LINAC area access. All sources of harmful radiation within the Vendor's proposed LINAC shall be identified and interlocked with the CLS ACIS system. The ACIS provides reliable, redundant, and failsafe dry contacts to interlock the radiation sources.

Every source of harmful radiation shall provide a minimum of 2 diverse interlocking points. For example, an RF source would incorporate separate RF power and RF trigger interlocks, and the gun would incorporate separate power (high voltage) and gun trigger interlocks. In addition, the gun shall include a separate interlock to cause a controlled rundown of the gun high voltage to zero.

Radiation output energy shall be reduced to zero when an interlock is activated. The interlocks shall be fail safe, meaning when the ACIS dry contacts are opened, the output energy is zero, and when the ACIS dry contacts are closed, LINAC operation is permitted. The transition of an ACIS interlock from opened to closed shall be made insufficient to turn the LINAC; an operator 'ON' command must be received after all interlocks are first closed.

The Vendor shall provide a safety rated control circuit for the ACIS to interface with, such as the Schneider-Electric XPSAF5130, which shall have a safety integrity level of (SIL) of 3 as per EN62061, or Performance Level (PL) 'e' as defined in ISO13849-1. Radiation hazards shall be reduced to zero within 100 ms of the activation of the interlocks, except in the case of the gun high voltage rundown interlock, which shall reduce the high voltage to zero within 60 s. For each of the diverse interlocks identified, the ACIS will provide two separate dry contacts.

Proof of compliance of safety performance rating shall be provided by the Vendor, including actual shutdown time.

5.13.2 Equipment cabinet doors shall interlock equipment contained inside if opening the door introduces an electrical or other hazard. If equipped, the cabinet door interlocks shall be failsafe and the state monitored by the controls system. Door interlocks shall be capable of being manually bypassed for the purpose of troubleshooting.

5.14 PROGRAMMABLE LOGIC CONTROLLERS

5.14.1 Machine Protection System

The Vendor shall provide a PLC based machine protection system (MPS) that will protect the equipment from damage and prevent operation until properly configured. The PLC shall be based on the Schneider-Electric M580 or the Siemens S7-1500. Alternative hardware is subject to CLS approval.

Typical functions of the MPS are to close vacuum valves in the case of poor vacuum and prevent generation of the electron beam unless vacuum valves are open, and equipment instrumentation readings are within safe operating range.

Maintaining good vacuum and limiting contamination from leaks is of particular importance to the CLS. The MPS PLC shall be configured to close 2 vacuum valves upstream and 2 downstream in the event of a poor vacuum reading. In addition, a fully separated "Vacuum PLC" shall be implemented, with the sole function of interlocking vacuum valves based on the vacuum instrumentation. The Vacuum PLC shall use different vacuum controller contacts than the primary MPS PLC.

The controllers for ion pumps, vacuum gauging (CCGs and TCGs), and other controllers required for implementation of the MPS PLC are in the Vendor scope of supply. Controllers shall have serial or Ethernet interface for direct remote communication.

Vacuum controllers shall be configured to prevent operation of the CCG when the TCG vacuum reading is poor. Vacuum gauging controllers shall provide at least two dry contact outputs per gauge, with separately adjustable trip points and hysteresis. CCG controllers shall provide adjustable self-protection.

The PLC shall retain the status of any device that interlocks operation of the LINAC and provide a latched indication to the operator control panel, alarm handler and archiver. The status shall remain latched until reset by the operator.

5.14.2 PLC Software Inputs And Outputs

All LINAC input and output data shall be accessible to the CLS controls system.

A computer callable interface to use EPICS compatible driver to change and read all necessary PLC parameters shall be provided.

This includes but is not limited to internal diagnostic monitor values, fault states, software enables, software disables, set points, feedback values, and software resets.

At minimum, the state of the following equipment shall be available on control screens:

Signal Description	Status	Comments
Power Supply	ON, OFF; RESET, INTERNAL FAULT, EXTERNAL FAULT, CURRENT or VOLTAGE	
Vacuum	Vacuum good interlock VVR closed, open TCG, CCG, IOP, RGA	
LCW	TM analog and interlock, FLT analog, PT analog, SWF interlock	All the cooling systems must generate temperature alarms when temperatures are over or under programmable thresholds.
External Interlocks from CLS as noted below		

5.14.3 PLC Hardware Inputs and Outputs to CLS equipment:

Outputs shall be provided for the vacuum status of every vacuum section of the LINAC. The status from the primary PLC and the vacuum PLC shall both be provided. These outputs shall be optically isolated dry contacts. Contacts shall be rated at minimum 24 VDC at 2 amps.

Inputs shall be suitable for interface to optically isolated dry contacts, rated at 24 VDC and 2 amps.

Inputs shall be provided for the vacuum status of the next two sections of the LTB transfer line, to both the primary PLC and the vacuum PLC, for a total of 4 inputs. The Vendor PLCs shall interlock vacuum valves as required.

8 inputs shall be provided to interlock generation of the LINAC electron beam. The CLS will provide output signals from systems such as the AARMS, LTB and BR.

Interlocks should be designed to be failsafe. A safe state shall be indicated by a closed contact sending a DC voltage. An unsafe state shall be indicated by an open contact that blocks the DC signal. On power failure the system should indicate an unsafe state.

When some of the interlocks have been tripped, the system or subsystem shall not be operational, even if the cause of the interlocks trip has been cleared, until the operator has reset the interlocks, either manually when in local mode or remotely when in remote mode. The error conditions must be identified both by the operator and the control system.

5.14.4 Remote Control System Interface

The control system PLC shall interface to the CLS control system PLC network through Ethernet using EPICS. All the control commands and read/write variables used in the local control system shall be visible over the CLS network. The LINAC shall run and protect itself autonomously regardless of the remote control system.

5.14.5 Panels

PLC panels shall not be more than 75% full. Panels shall allow for future addition of input and output modules and associated terminal blocks.

5.14.6 Source Code

The Vendor shall provide all PLC source code, in English, to the CLS.

5.15 ELECTRICAL SERVICES

5.15.1 Power

All equipment requiring 3 phase power shall be suitable for use with 480 or 208 volts line-to-line supply. Three phase equipment larger than 7.5 kW (10 hp) shall utilize a 480 volt supply.

All equipment requiring single phase power shall be suitable for use with 277 or 208 or 120 volt supply. Single phase equipment smaller than 750 W should utilize a 120 volt supply.

All equipment such as Motors, electric heating, control and distribution devices and equipment to operate satisfactorily at 60 Hz within normal operating limits established above.

Equipment shall operate on voltage tolerances +/- 10% to the above values.

Equipment to operate in extreme operating conditions established above without damage to equipment.

5.15.2 Arc Flash Study

CLS shall supply arc-flash labels on each distribution board and panel board supplying power to the vendor's supplied LINAC equipment. Vendor shall provide breaker, fuse and transformer information to CLS for coordination and arc flash power study.

5.15.3 Connection

Equipment shall include plugs compatible with the CSA receptacle configurations from Diagram 1 or Diagram 2 in the Canadian Electric Code, where intended to be detachably connected.

5.15.4 Power Interrupt Current

Where supplied, fuses and MCCBs shall have interrupting currents rated at 42000 A.I.C. or higher at 480VAC.

Controls power interrupt current shall be 10000 A.I.C. or higher at 208VAC.

5.15.5 Safety Disconnect

The electrical equipment shall have an integral safety disconnect that safely disconnects the AC power source for the load. This disconnect shall be lockable in ONLY the AC power OFF position.

5.15.6 Safeguarding

Equipment shall include safeguarding to prevent persons from coming into contact with hazards, or reduce hazards to a safe state, before a person can come into contact with them. Emergency stop devices shall comply with EN 60947-5-5.

Magnets will be constructed in accordance with best practice. The coil terminals, the connection posts and all metallic parts connected to them should be protected against accidental contact by an insulating transparent cover.

5.15.7 Grounding

All portions of the LINAC, including input and output connections, shall be at the same ground potential.

5.15.8 Ground Currents:

The Manufacturer shall design a system with negligible ground currents.

5.15.9 AC Mains Transients

The Vendor shall anticipate and take precautions against LINAC damage due to incoming AC mains transients.

5.15.10 Shielding

All voltages (greater than 50 V) shall be shielded. High voltage components, connectors, wiring terminations, etc. shall be physically separated from low voltage control circuits. Control signals, interlocks, and diagnostic signals shall be shielded and routed for noise immunity.

6.0 SAFETY PERFORMANCE

6.1 OPERATING ENVIRONMENT

- 6.1.1** All components shall be designed to operate in a radiation environment. The manufacturer shall avoid using materials that are subject to damage by ionizing radiation or provide adequate shielding incorporated into the design to allow for long service life of the component. All components should have a minimum mean time between failures of 10 years in the typical radiation environment.

7.0 WORKPLACE SAFETY AND ENVIRONMENT

- 7.1.1** All employees and subcontractors of the vendor shall conform to all CLSI safety policies and procedures. These include, but are not limited, to forklift operation, scissor lift operation, fall arrest, hot work and fire extinguishers, tool usage and facility access requirements. Vendor shall provide copies of Proof of Training for review by CLSI and may be required to provide demonstration of competency prior to operating equipment on site.

- 7.1.2** The components shall be able to withstand a relative humidity range of 0% to 90% noncondensing. The expected relative humidity limits under operation are from 25% during the winter months to 50% during the summer months. Equipment shall be suitable for use at an elevation of 480 meters.
- 7.1.3** Saskatoon, Saskatchewan experiences a wide range of temperature and weather events. This information is easily found on the internet and should be reviewed for the anticipated times of deliveries and installation.
- 7.1.4** CLS is a 24 hour, 7 day a week, 365 days a year operating facility. There are scheduled maintenance times when operations are suspended. Normal CLS office hours are Monday to Friday, 8:00 AM - 4:30 PM.

8.0 APPLICABLE CODES, STANDARDS, AND PROCEDURES

This work shall meet the following standards. The issue of any standard shall be the issue in effect as of the date of request for tender. Any conflicts between this specification and the referenced documents shall be brought to the attention of the CLS in writing for resolution before any related action is to be taken by the Vendor.

- American Welding Society (AWS).
- American Society for Testing and Material (ASTM).
- CSA B51-97 Boiler, Pressure Vessel, and Pressure Piping Code.
- CSA B52-99 Mechanical Refrigeration Code.
- Canadian Light Source Vacuum Component Cleaning Technical Procedure [5].
- Canadian Light Source Vacuum Component Leak Test Technical Procedure [7].
- Canadian Light Source Inc. Vacuum Design Specification [3].
- Canadian Light Source Vacuum Component Fabrication Technical Specification [4].
- Control System Common Specification [2].
- C22.1-18 Canadian Electrical Code, Part I (2018), Canadian Standards Association
- Saskatchewan Interpretations (2015), SaskPower
<https://www.saskpower.com/-/media/saskpower/accounts-and-services/service-requests/guide-electricalcode-saskpower-interpretations-2015.ashx>
- Saskatchewan Electrical Inspection Act, E6-3
www.qp.gov.sk.ca/documents/English/Statutes/Statutes/E6-3.pdf
- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment; MIL-STD-461F, 10 DEC 2007
- IEEE Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz; IEEE C95.1, 2005.
- Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems; IEEE Std 519-2014
- Industrial, Scientific and Medical Radio Frequency Generators, Industry Canada ICES-001

- IEEE Std C57 Series for Distribution, Power and Regulating Transformers including but not limited to: C57.12.01, C57.12.56, C57.12.59, C57.12.80, C57.12.91, C57.110
- Air-Cooled Transformers (Dry Type); CSA C22.2 No. 47-13
- Low-Voltage AC Power Circuit Breakers Used in Enclosures - Test Procedures; ANSI C37.50

9.0 QUALITY

The vendor shall maintain and apply a quality assurance program for the design, manufacture and testing. This program shall be submitted to CLS for review and acceptance. The CLS and the vendor shall work together to monitor the progress of the project and any problems and delays on either side shall be promptly communicated between the parties.

10.0 DESIGN REVIEW, INSPECTION, TESTING AND COMMISSIONING

The vendor shall prepare an Inspection and Test Plan (ITP) for all major pieces of equipment or systems within the vendor's scope of supply and submit this test plan to CLS for review and approval.

The ITP should include at minimum each of the following milestones (section 10.1 to 10.4) together with the documents and activities required at each milestone.

10.1 PRELIMINARY DESIGN REVIEW

10.1.1 Preliminary General Arrangement Drawings

10.1.2 Preliminary Electrical & Controls design

10.1.3 Preliminary Mechanical design

10.1.4 Preliminary component selection

10.1.5 Preliminary designs shall be agreed upon by CLS and Vendor.

10.1.6 Preliminary 2D and 3D drawings shall be provided to CLS. They shall be subject to CLS approval.

10.2 FINAL DESIGN REVIEW

10.2.1 Final General Arrangement Drawings

10.2.2 Final Electrical & Controls design

10.2.3 Final Mechanical design

10.2.4 Finalize component selection, including type and model of components and the basic system layout

10.2.5 Final designs are subject to the approved of CLS

10.2.6 Final 2D and 3D drawings shall be provided to CLS. They shall be subject to CLS approval.

10.2.7 CLS will have the right to access and check the calculations and simulations used as input to the designs, fully respecting the IP of the vendor and CLS and the vendor will make use of NDAs where required.

10.3 FACTORY INSPECTION AND FACTORY ACCEPTANCE TESTING

10.3.1 List of all equipment subject to factory inspection and factory acceptance testing.

10.3.2 Type of inspection and acceptance test

10.3.3 Vendor witness and hold points

10.3.4 CLS witness and hold points

10.3.5 Schedule for submittal of detailed inspection and test documents in advance of test.

10.3.6 Schedule for actual inspection and tests.

10.3.7 An option shall be provided by the vendor to complete high power RF conditioning of the accelerating structures to minimize the commissioning time after installation at the CLS.

10.3.8 An option shall be provided to test and condition the modulator and klystron systems at the CLS prior to installation to minimize the commissioning time.

10.3.9 CLS may require a representative to observe measurements that are part of the inspection and acceptance tests. Notice of the start of the factory inspection or acceptance tests must be given 3 weeks prior to starting the tests to allow for travel arrangements.

10.4 CLS SITE ACCEPTANCE TESTS

10.4.1 The vendor shall propose a site acceptance test (SAT). The CLS site test plan shall reviewed and accepted by CLS. They shall include the items in this Section, 5.3.

10.4.2 The vendor control system tests including operational tests of the motors, limit switches, encoder calibrations, and operation of safety switches and contacts.

10.4.3 Vacuum system performance testing and verification, including helium leak check, verification of vacuum pressure after bake out, and RGA analysis vacuum system.

10.4.4 A test of the water cooling system at a pressure of 1.5 times operating pressure ($1.07 \text{ MPa} \times 1.5 = 1.6 \text{ MPa}$) for 8 hours. If the test pressure would produce a nominal pressure stress or longitudinal stress in excess of the yield strength at test temperature, the test pressure may be reduced to the maximum pressure that will not exceed the yield strength at test temperature, as stated in ASME B31.3-2008 paragraph 345.2.1.

11.0 CALIBRATION

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12.0 RELIABILITY AND MAINTAINABILITY

The expected life time of the Electron Source and LINAC and all sub-components shall be greater than 20 years. Systems shall be designed and constructed for continuous use with high reliability and stability.

The vendor shall provide a recommended maintenance schedule for all equipment supplied.

The vendor shall provide a list of recommended spare parts complete with part numbers, quantity and description. The list shall include any spare parts that are critical for operation and parts subject to wear and requiring routine replacement or maintenance.

13.0 OTHER REQUIREMENTS AND CONSTRAINTS

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14.0 DRAWINGS

The following drawings are supplied as part of his technical specification.

- LINAC ROOMS AND ACCESS AREAS FOR LINAC UPGRADE (2 sheets); CLS Drawing ACCL/PPL/0047250 REV.0
- MODULATOR ROOM LAYOUT FOR LINAC UPGRADE; CLS Drawing ACCL/PPL/0047251 REV.0

15.0 DOCUMENTATION

The vendor shall supply the documentation specified within the other sections of this specification in addition to the following;

15.1.1 Initial meeting documentation and minutes following contract award.

15.1.2 A schedule for the manufacturing, inspection, delivery, installation, commissioning and submittal of documents. This schedule shall include the estimated CLS manpower requirements for installation and commissioning.

15.1.3 An Inspection and Test Plan for all equipment being supplied.

15.1.4 Monthly progress reports.

15.1.5 Preliminary design review document.

15.1.6 Final design review document.

15.1.7 All test reports.

- 15.1.8** Operating and Maintenance Manuals for other supplier's components and the vendor's LINAC as a system.
- 15.1.9** Full 2D and 3D insertion device mechanical and electrical design drawings in dwg or step format.
- 15.1.10** Documentation requirements are further defined in CLS document, 0.4.1.1, Vendor Documentation Specification with the following exceptions that supersede.
- 15.1.10.1** CLS is currently using Autodesk Inventor 2016 and AutoCAD 2016 CAD software. Altium Designer 17.1.6 is the current PCB layout software.
- 15.1.10.2** Autodesk AutoCAD Electrical, CadSoft Eagle and Microsoft Visio are not used.

16.0 REFERENCES

- [1] Canadian Light Source Requirements for Design and Fabrication of Structural Supports; CLS Document 8.8.29.1
- [2] Control System Common Specification; CLS Document 7.4.39.1
- [3] Canadian Light Source Inc. Vacuum Design Specification; CLS Document 8.4.33.1
- [4] Canadian Light Source Vacuum Component Fabrication Technical Specification 8.8.33.4
- [5] Canadian Light Source Vacuum Component Cleaning Technical Procedure; CLS Document 8.7.33.1
- [6] Design Specification for Precision Aligned Components; CLS Document 0.4.1.2
- [7] Canadian Light Source Vacuum Component Leak Test Technical Procedure; CLS Document 8.7.33.2
- [8] Canadian Light Source Vendor Documentation Specification; CLS Document 0.4.1.1
- [9] CLS Naming and Numbering Convention; CLS Document 0.2.1.1

REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	2018-Nov-21	Original Draft	Gordon Barkway
B	2018-Nov-23	Incorporated comments from reviewers	Gordon Barkway
0	2018-Nov-23	Issued for use	Gordon Barkway